

CSC1118 Enterprise Computing Team Project

Project Title: PulsePath

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Mid-Term Delivery

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1. Introduction

1.1. Overview

PulsePath is a web-based solution designed to enhance festival crowd management through real-time insights. It provides festival organisers with tools to monitor crowd density, optimise resource allocation, and improve overall safety and efficiency at medium to large-scale festivals.

1.2. Context

Festival organisers face challenges such as high security costs, overcrowding risks, and inefficient resource allocation, which can compromise attendee safety and event success. PulsePath addresses these issues by offering a cost-effective platform that combines real-time monitoring and actionable insights to streamline operations and enhance safety.

1.3. Document Scope

This document outlines the initial business model for PulsePath, including its concept, value propositions, target audience, and revenue streams. It also details the project's functional specification, software architecture, key challenges, and proposed timeline, supported by an appendix for additional content and evidence.

2. Business Model Summary

2.1. Concept

The concept behind PulsePath's business model is to provide festival organisers with an innovative, cost-effective solution to the persistent challenges of crowd safety and resource management. Traditional methods often rely on costly security teams and manual oversight, leaving organisers struggling to manage overcrowding and optimise resource allocation effectively. As noted in the 'Traditional Methods' section of Appendix A, Lisa Jameson highlighted the reliance on manual clickers for tracking attendees, emphasising the limitations of traditional approaches. PulsePath offers a streamlined alternative through real-time crowd density monitoring, actionable insights, and resource planning tools. This business model focuses on delivering measurable value to organisers by reducing safety and crowd control costs, improving operational efficiency, and enhancing attendee experiences, all while maintaining a scalable and flexible pricing structure. The following section breaks down how PulsePath's concept is structured to address these critical needs and create long-term value for its customers.

2.2. Value Proposition

PulsePath's value proposition is simple: optimised safety with smarter spending. By providing festival organisers with real-time crowd monitoring, resource allocation tools, and actionable insights, PulsePath aims to reduce safety costs by up to 20% while improving operational efficiency. With its user-friendly interface and customisable features, the platform empowers organisers to manage crowds effectively, minimise risks, and deliver better experiences for attendees, without the need for excessive security teams or outdated methods. PulsePath offers a smarter, safer, and more cost-effective way to manage festivals.

2.3. Target Audience

PulsePath's target audience is medium to large-scale festival organisers who face ongoing challenges with crowd management, safety, and resource allocation. These organisers are responsible for ensuring the safety and enjoyment of thousands of attendees while balancing tight budgets and operational pressures. By offering tools that provide real-time crowd insights and reduce reliance on costly security measures, PulsePath caters to organisers looking for practical, cost-effective solutions to improve event safety and efficiency. This audience includes music festival managers, cultural event planners, and large-scale outdoor event organisers who are eager to enhance safety standards and streamline operations.

2.4. Market Size

PulsePath focuses on Ireland's growing market for crowd control and safety technology, offering innovative solutions that combine real-time crowd insights with predictive analytics. The total addressable market for crowd control and safety technology in Irish music events is projected at €13.35 million in 2024, with the serviceable addressable market focusing on cloud-based solutions valued at €8.01 million (see Appendix B for detailed calculations). PulsePath's distinct value proposition positions it to capture 20% of this SAM, equating to a serviceable obtainable market of €1.60 million in its first year (refer to Appendix B, SOM Calculation).

Our market size estimation is supported by primary research conducted with Lisa Jameson, Events Manager at Teeling Whiskey. Jameson highlighted the challenges of managing crowd behaviour and resource allocation, emphasising the importance of real-time observation, staff adaptability, and feedback-driven improvements. She noted the unpredictability of crowd dynamics and the need for contingency plans, reinforcing the demand for solutions like PulsePath that provide actionable insights. For example, managing bottlenecks at bars and entry points or adapting to larger than expected attendance, as seen during their outdoor events, aligns with PulsePath's ability to dynamically monitor crowd density and predict potential issues (refer to Incident Management and Contingency Planning, Appendix A).

Jameson mentioned budget constraints often shape decisions about safety measures, further validating PulsePath's potential to reduce safety costs by up to 20% through better resource allocation. With PulsePath's pay-per-use pricing model, event organisers can enhance safety without long-term financial commitments, addressing a critical need in the industry. This combination of industry trends, practical insights, and our market analysis creates a clear path for PulsePath to capture a significant part of the €1.60 million obtainable market in 2024.

2.5. Revenue Streams

PulsePath generates revenue through three primary streams, each designed for reliability and growth based on market insights. The primary stream is a pay-per-use fee, which aligns with industry trends in SaaS pricing and offers organisers flexibility without long-term commitments, making it appealing to cost-conscious customers (see Appendix E for pricing strategy analysis). A premium post-event analytics package provides organisers with actionable insights, such as crowd flow and safety optimisation, addressing the growing demand for data-driven decision-making in event planning (see Appendix D for detailed insights). While this feature will not be included in our final deliverables due to resource and time constraints, our research highlights its potential. Market analysis from the Irish festival sector shows an expanding interest for these insights, with a projected total addressable market of up to €28.68 million by 2028. In-app advertising from food vendors and businesses leverages a targeted audience of festival attendees, offering vendors significant ROI through increased visibility (see Appendix C for ROI calculation). These revenue streams are supported by industry trends, demand, and scalable opportunities, ensuring a strong business model for PulsePath.

3. Summary Functional Specification

3.1. General Description

Our aim with PulsePath is to create an innovative real-time crowd management system that will be designed in a way so that it can be utmost beneficial for open-air festivals. By leveraging location tracking and data visualisation technologies, PulsePath provides festival organisers with dynamic insights into crowd densities and movements, enhancing safety and improving event management. The main purpose of the web app is to enhance the safety of individuals on the festival grounds and simultaneously improve the attendee experience by managing crowd-related risks and contributing towards the optimisation of resource distribution within the festival grounds. Our system aims to utilise live location services in order to monitor the crowd flow and density, which in return will help prevent overcrowding and manage emergency situations amidst large crowds more effectively.

3.2. Key Functional Requirements

3.2.1. Interactive Festival Map

The Interactive Festival Map is a pivotal feature of PulsePath. This feature will be designed to enhance both navigation and safety at open-air festivals. Utilising the advanced mapping capabilities of MapBox, the map will be overlaid with the actual festival layout, including accurate representations of stages, toilets, food stalls, and other essential amenities.

Each amenity on the map is surrounded by virtual boundaries known as geofences. These geofences are designed to detect the entry and exit of festival-goers' devices through GPS. As attendees move throughout the festival, their entrance into or exit from a geofence is automatically detected. Amenities will also be equipped with a dynamic text bubble. As people enter and exit the geofences, their movement will trigger an update in the count displayed on the corresponding bubble on the map.

This feature helps in monitoring and managing crowd levels at various points of interest within the festival grounds.

For example, when a festival-goer approaches and enters a geofence around a food stall, the system recognises this entry. The count on the bubble associated with that food stall increments by one, reflecting the new total of people within that specific geofence. Similarly, when an individual leaves the geofence, such as moving away from a stage after a performance, the count on the bubble decreases accordingly.

The text bubbles or their backgrounds are colour-coded to reflect the capacity load—green indicates low capacity, orange signals medium capacity, and red indicates that the area has reached or is near maximum capacity. This intuitive colour-coding system allows festival-goers and organisers to visually assess crowd distribution at a glance.

This feature is very helpful for real-time crowd management, as it allows both attendees and organisers to see up-to-date information on how crowded specific areas are.

As well as getting information on crowd capacities, attendees can interact with the map to explore different amenities and stages, plan their routes, and avoid overcrowded areas. This level of interaction not only improves the user experience but also contributes to a safer festival environment by contributing towards the prevention of potential bottlenecks and improving the flow of foot traffic.

3.2.2. Personal Line-up Feature

The Personal Line-Up Feature of PulsePath transforms the attendees' festival experience by enabling them to create a customised schedule that aligns with their interests. This feature leverages the app's database of artists and event timings, allowing users to tailor their day to ensure they don't miss their favourite performances.

Users begin by browsing the full list of performances scheduled for the festival. Each artist or act is listed with details about their performance time and stage location. Users can select to 'heart' their favourite artists or acts by tapping on the heart icon next to each name. Once users have selected their favourites, the app automatically generates a personalised line-up. This line-up includes only the chosen performances, organised by time and location. The app ensures that there are no time conflicts between selected performances, and if any overlap occurs, users are prompted to decide which act they prefer to see.

3.2.3. Real-Time Updates

We aim to integrate real-time push notifications to the users' browsers even when they aren't actively using the web app. This feature is designed to keep users informed about critical changes and updates that affect their planned activities and preferences.

Users who have selected their favourite artists through the Personal Line-Up Feature will receive timely notifications just before their set times. This ensures that users can make their way to the correct stage in advance, enhancing their experience by never missing a performance they care about.

If there are any last-minute changes to the performance times or stages of favored artists, users will receive an immediate alert. These notifications help users adjust their schedule on the fly, maintaining the fluidity of their festival experience without the hassle of constantly checking for updates manually.

Users can opt to receive notifications about any changes in the crowd capacity levels of the stages or areas they are interested in. This feature can come in handy when helping attendees decide when they should move from one stage to the next, as they can get alerted of the capacity of a certain stage increasing. For users that choose to receive notifications on the live capacity alerts of stages/amenities they will be informed once a certain stage/amenity has changed status, for example they will be alerted when the capacity has changed from green (low) to medium (orange).

3.3. Software Architecture

The software architecture of PulsePath is designed to support a robust, scalable, and efficient real-time web application capable of handling the dynamic needs of festival crowd management. Our architecture is built on a modern technology stack that emphasises responsiveness, user engagement, and real-time data processing.

Frontend: Our frontend will be developed using HTML5, CSS, Javascript with React.js in order to create a dynamic and responsive user interface. This setup allows us to integrate a map, live updates and personalised schedules to operate fluidly.

Backend: The backend will be powered by Django. Django runs on a web server that interfaces with our application to handle client requests, manage sessions and deliver content. The application logic, including user authentication, data processing, and notification dispatch, is handled by Django. Django's built-in components are used for ORM (Object-Relational Mapping), URL routing, and session management. [1]

Database: For our database we chose PostgreSQL due to its robustness and compatibility with Django. It supports our complex queries and data requirements, especially for storing artist schedules, geofence coordinates and real-time crowd metrics.

Real-Time Processing: In order to achieve real-time communication between the client and the server we're going to integrate WebSocket. WebSocket will help us with pushing updates to the frontend as they happen. This will be extremely useful when it comes to changes in crowd density or updates to the event schedule.

External APIs: MapBox API provides a suite of powerful endpoints that will allow PulsePath to create customised, scalable and interactive maps. MapBox APIs are integral to PulsePath due to their ability to handle complex geographic data and render it in a user-friendly manner. The APIs support scalable solutions, ensuring that as the festival grows in size or scope, the app can handle increased demands without compromising performance. Moreover, the real-time capabilities of MapBox APIs are crucial for live event management, where conditions change rapidly, and up-to-date information is vital for operational success.

Using the MapBox Styles API, PulsePath will design and implement custom maps tailored specifically to each festival layout. This allows us to incorporate unique design elements that align with the festival's theme and provide a more intuitive user experience.

The MapBox Vector Tiles and Raster Tiles APIs will be used to overlay real-time data on the maps, such as crowd densities and the locations of various amenities. This feature is crucial for providing live updates that help in managing crowd control and enhancing safety.

The MapBox Geocoding API will be utilised to convert locations into coordinates and vice versa. This is particularly useful for features like finding nearby hotels or shuttle services, enhancing the overall user experience by integrating broader geographic information. [2]

Session Management and Caching: Effectively managing session data and caching is crucial to ensure a seamless user experience, especially during high-traffic events such as the festivals we aim to be working with. We plan on using Redis as a session store to manage user sessions throughout the application. This includes possibly storing user preferences, login states and personal line-ups. Redis can also handle a large number of connections simultaneously, which is crucial during peak festival times when thousands of users may be interacting with the app simultaneously. By storing session data in Redis, the load on the main database is reduced, thereby enhancing overall performance.

Choosing Redis also gives us the flexibility to use its built-in mechanisms for setting expiration on stored data. This is particularly useful for session management where data only needs to persist the duration of the user's activity. Sessions can be configured to expire automatically after a set period of inactivity and this will help us manage storage, ensuring old user data does not consume unnecessary resources.

We will also be able to utilise Redis to cache frequently accessed data such as the festival line-ups and stage schedules. This will in return reduce the direct database queries needed, thus speeding up the response times for user requests.

Redis is also easy to integrate with our Django backend through the use of libraries such as *django-redis*, which facilitates an easy setup and management of caches and session data. [3]

Scheduled Jobs and Background Tasks: To effectively manage asynchronous tasks and background processing within our web app, we will utilise Celery, a powerful distributed task queue system. Here's how we plan to implement and leverage Celery in our system. We will be able to integrate celery with our Django backend, making it easier for us to configure and monitor tasks directly from the Django admin panels. As Celery requires a message broker to send and receive messages, Redis will do this for us since it is compatible with Celery.

We plan to mainly use Celery in order to send to batch notifications or process other data for analytics without having to run anything in the immediate path of user requests. We are also able to make use of Celery Beat, a scheduler for Celery, that we can use to run scheduled tasks at regular intervals. For example, updates to the festival line up or countdown notifications.

Another benefit of using Celery is that it allows for task prioritisation. Critical tasks such as our real-time alerts on crowd capacity can be prioritised over less critical tasks like sending promotional notifications. [4]

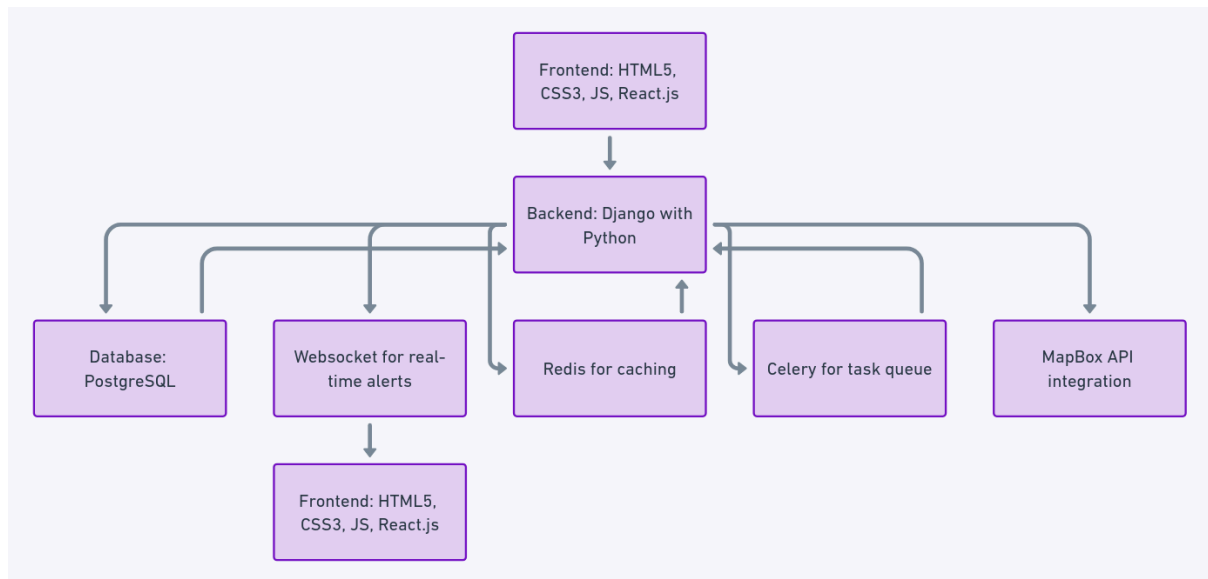


Figure 1.1 Software Architecture Diagram

3.4. Prototype Scope

The primary objective of our prototype is to demonstrate the core functionalities that will address the essential needs of festival organisers and attendees. These include the interactive map, real-time notifications and personal line-up feature. The prototype will serve as proof of the concept that our web app will enhance the festival experience through our innovative technology solutions.

Our aim is for the prototype to include a basic version of the Interactive Festival Map using the MapBox API. This will allow users to view the layout of a hypothetical festival site, including key locations such as stages, food stalls and restrooms. Through this prototype we want to test the map's responsiveness, usability and accuracy in a simulated environment.

The prototype will include the real-time notification system that alerts users about events such as the start of an act or changes in crowd density at specific areas. Here our aim is to validate the system's capability to deliver timely and reliable notifications under various network conditions.

Additionally, our prototype will include the personal line-up feature where users will be able to select their favourite artists from the line-up and create a personalised schedule. This feature is all about giving people an easy way to organise their day and focus on what matters most to them.

By focusing on these features, our prototype will serve as the first step in showing how our app can improve the festival experience for both organisers and attendees.

3.5. Challenges

Addressing potential challenges in a project is critical for planning and mitigation strategies. One of the main challenges we anticipate to phase is with integrating the geofence logic and the technical challenges that will present for our project. This functionality is central to our objective of enhancing the real-time crowd management during festivals and therefore ensuring its functionality is our main concern.

Our aim is to combine GPS and Bluetooth technologies in order to manage the geofences. This involves coordinating two different types of data inputs to trigger location-based actions. GPS can provide broad location tracking whilst Bluetooth can detect proximities more accurately within small areas. However, ensuring seamless integration and data synchronisation between these technologies is challenging and requires complex software logic to manage potential discrepancies in data accuracy and timeliness.

Additionally, utilising Bluetooth technology in order to count nearby devices assumes all festival-goers have Bluetooth enabled and are discoverable, which may not always be the case. The approach also relies on at least a few devices within each geofence having a stable internet connection to communicate with the server. Inaccuracies in counting or failure to transmit data due to connectivity issues could lead to unreliable crowd density data, affecting decision-making for crowd control.

Our mitigation strategy to dissolve these issues include developing algorithms that will intelligently combine GPS and Bluetooth data to optimise accuracy and reduce dependency on any single technology whilst also encouraging users to enable Bluetooth and explain the benefits and privacy measures in place to assure them of their data's security.

Addressing these challenges with innovative technical solutions and thorough testing is crucial for the successful implementation of PulsePath's geofencing feature. Our project supervisor is optimistic that with our strategic approach and continuous effort, we will be able to overcome these challenges and achieve our project goals effectively and this encourages us to strive towards a better result.

4. Project Timeline

4.1. Milestones

The project timeline for PulsePath focuses on delivering a web application by completing specific technical tasks and using effective project management strategies. Below are the milestones with details on how tasks will be achieved:

1. Customisation and Core Map Features (End of January 2025):
 - Customise the MapBox map by designing and uploading custom icons for festival facilities using the MapBox Studio interface.

- Begin setting up WebSocket connections to stream real-time updates from the backend to the map interface. Progress will be tracked using a JIRA board, with frequent online and in-person check-ins.
2. Backend Development and Feature Integration (February 2025):
 - Develop backend APIs in Django to handle geolocation updates, user requests, and crowd density data. These APIs will interact with our PostgreSQL database to store and retrieve information efficiently.
 - Use WebSocket protocols to establish real-time communication between the backend and front-end, ensuring updates to the map are instantaneous.
 - Integrate these backend features with the MapBox-based front-end. Frequent meetings with the project supervisor will give us the opportunity to discuss any technical challenges we are facing.
 3. Testing and Functional Refinement (March 2025):
 - Conduct integration testing by simulating crowd density scenarios using test data, verifying that the WebSocket updates are accurately reflected on the map.
 - Test geolocation accuracy by creating sample location datasets and validating their visual representation on the map.
 - Perform usability testing, involving peers to interact with the interface, gathering feedback to refine user interactions and design elements. Progress will be recorded in the JIRA board to maintain task clarity.
 4. Final Development (Early April 2025):
 - Conduct stress testing of the WebSocket communication by simulating high loads of concurrent location updates to ensure the system remains stable.
 - Debug and fine-tune all application features, including notifications and real-time updates by reviewing logs.
 - Complete internal documentation for the codebase and application features, ensuring future maintainability.
 5. Final Deliverables Preparation (By 18th April 2025, 17:00):
 - Organise and prepare all project materials, including source code, the final functional specification document, and user documentation.
 - Ensure all files are uploaded to the project repository in the required formats before the submission deadline.
 6. Presentation and Expo Preparation (Late April to Early May 2025):

- Prepare a technical presentation demonstrating the implementation process, focusing on key features like real-time updates and map customisation.
- Create supplementary materials such as a demonstration script and visual aids for the Project Expo.
- Conduct a mock presentation with our project supervisor to identify areas for improvement.

7. Final Presentation and Expo (5th to 16th May 2025):

- Use the completed application to showcase PulsePath's core functionalities, presenting crowd density monitoring and map interactivity, during group interviews and the Project Expo.

Throughout the project, the team will use a JIRA board to monitor task progress and ensure deadlines are met. Regular online and in-person meetings, along with consistent feedback from the project supervisor, will support progress and resolve challenges effectively.

4.2. Roles and Responsibilities

We have divided our responsibilities to ensure smooth progress and effective collaboration. I (Zeynep) have completed foundational tasks such as integrating MapBox for interactive mapping features and setting up the PostgreSQL database to provide a strong backbone for data management. Lizel has worked on organising the project folder and subfolder structure, ensuring that the Django framework is correctly configured to communicate with future technologies, such as WebSocket, while maintaining flexibility for further integration.

Together, we share the responsibility of documenting every step of our progress. This documentation is essential for tracking completed tasks and ensuring that we have clear records of how each task was executed. By updating our JIRA boards phase by phase, we follow an Agile methodology, which allows us to address any issues or inefficiencies early. This approach helps us stay on track to meet our milestones and adhere to the deadlines set by both our team and DCU.

From a technical perspective, Lizel focuses on backend development, including database management and server-side logic, while I take the lead on frontend architecture and ensuring compatibility between Django and the technologies we plan to integrate. Tasks like testing, debugging, and ensuring overall system compatibility are joint responsibilities, requiring both of us to collaborate closely. By leveraging our strengths and working together on shared tasks, we aim to deliver a functional PulsePath system within our proposed timeline.

4.3. Gantt Chart

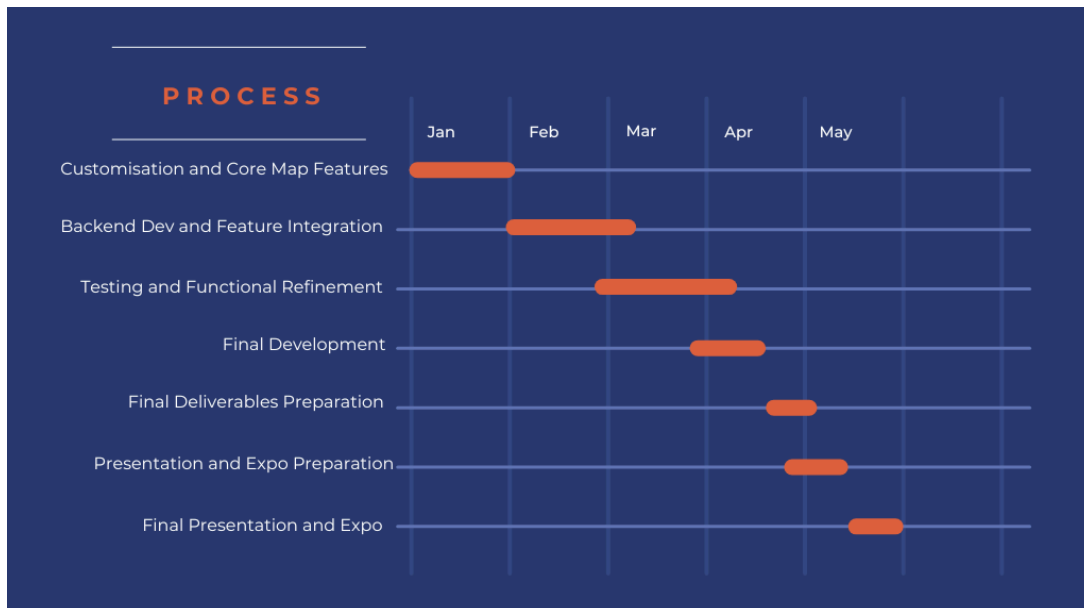


Figure 1.2 Process Gantt Chart

5. Conclusion

PulsePath represents an innovative step forward in festival crowd management, addressing critical challenges such as safety, resource allocation, and attendee experience. By leveraging advanced technologies like geofencing, real-time updates, and customisable tools, the platform offers a comprehensive solution tailored to the unique demands of medium to large-scale festivals.

Through its scalable business model, PulsePath aims to significantly reduce operational costs while enhancing safety measures, thereby empowering event organisers to focus on delivering memorable and secure experiences. The integration of interactive features, such as real-time crowd insights and personalised attendee line-ups, further ensures its usability and appeal to both organisers and participants.

Despite anticipated challenges in geofencing and technology adoption, the proactive strategies outlined in this report position the project for successful implementation. With a strong foundation in market research and technical planning, PulsePath aims to make a substantial impact on the Irish festival market, setting the stage for future expansion and innovation in event management technology.

6. References

[1] Jain, K. (2021) 'When to Use Django (And When Not To)', *Crowdbotics*, 6 October. Available at: <https://crowdbotics.com/posts/blog/when-to-use-django-and-when-not-to/>

[2] Mapbox (n.d.) 'API Documentation', *Mapbox*. Available at: <https://docs.mapbox.com/api/overview/>

[3] Redis Ltd. (n.d.) 'Redis Documentation', *Redis*. Available at: <https://redis.io/docs/latest/>

[4] Celery Project (2024) 'Celery - Distributed Task Queue', *Celery 5.4.0 documentation*, 30 September. Available at: <https://docs.celeryq.dev/en/stable/>

7. Appendices

Appendix A

Interview Transcript with Lisa Jameson, Events Manager at Teeling Whiskey

This interview was conducted to gain insights into the challenges and strategies involved in managing crowd behaviour and resource allocation during events. Lisa Jameson shared her experiences managing events of various scales at Teeling Whiskey Distillery, highlighting the need for adaptability, observation, and data-driven solutions.

Event Dynamics:

Question: Can you describe the typical size and scale of events hosted by the Teeling Whiskey Distillery? How do you plan for crowd control for varying attendance sizes?

Response: At Teeling Whiskey Distillery, most events are capped at 200 attendees, although some larger events, such as the tree lighting ceremony, can accommodate up to 400 people when held outdoors. Outdoor events bring unique challenges, particularly when dealing with larger crowds. For smaller events, up to 250 attendees, the priority is ensuring guest comfort rather than maximising capacity. We emphasise comprehensive evacuation training for our team, assigning staff to each exit and ensuring they remain calm and confident to avoid causing panic. For larger events, managing crowd flow through a single entry point can be a challenge, but we mitigate this with clearly defined roles and multiple fire exits (approximately six). Attendees are also briefed on their nearest exits, particularly when meetings are held upstairs.

Event Challenges:

Question: What are the most common challenges you face regarding safety and security during events?

Response: Serving alcohol introduces specific risks, including the potential for disruptive behaviour. It's crucial to identify and address concerns before they escalate. Crowd behaviour can vary significantly depending on the type of event. For corporate events, attendees are usually professional and well-behaved, but private events often require greater mindfulness as there is no designated individual in charge. Security teams monitor for issues such as external alcohol brought by guests and address challenges like impatience among coach groups or bottlenecks at entry points and bars. Observation and learning from past incidents are key, and we use guest feedback to refine procedures and improve the experience.

Staff Deployment:

Question: How do you determine the number of security personnel required for an event? Are there specific factors or metrics you consider?

Response: For events with up to 70 attendees, we rely on in-house security. Beyond this number, we bring in additional personnel, scaling security proportionally with attendance. For example, one extra staff member is added for every additional 70–100 guests. After events, we conduct debriefs and compile reports to evaluate what worked and what didn't, ensuring we adapt to address any unforeseen challenges.

Incident Management and Contingency Planning:

Question: Can you share an example of how you managed a safety or security incident at a past event? What worked well, and what could have been improved?

Response: During a tree lighting event, we anticipated larger-than-usual attendance without an RSVP system. We brought in additional safety measures, including fire brigade personnel and crowd control barriers, to prepare for potential overcrowding. Despite the unpredictability, our contingency plans ensured a smooth experience. When bottlenecks formed at bars, we opened additional temporary bars to alleviate pressure. This adaptability is key, and we continuously refine our approach based on observations and feedback.

Traditional Methods:

Question: How do you distribute security during events with multiple attractions?

Response: For the tree lighting event, we relied on data from previous years to estimate attendance, but without an RSVP system, we had to prepare for the unexpected. We used a manual clicker device to track the number of people entering and exiting, which helped manage capacity in indoor areas such as the café and exhibition space. This method worked better for indoor settings, but we remained prepared to adapt our approach as needed. Clickers are always part of our contingency planning to ensure safe and efficient crowd management.

Appendix B

Market Size Calculations

This appendix provides a detailed explanation of the calculations used to determine the market size for PulsePath's value proposition, focusing on optimised safety with smarter spending. The calculations cover the Total Addressable Market (TAM), Serviceable Addressable Market (SAM), and Serviceable Obtainable Market (SOM) for the Irish crowd control and safety technology market in 2024 and 2028.

Total Addressable Market (TAM): PulsePath operates within the growing market for crowd control and safety technology in the Irish music events sector. Based on the overall projected revenue for the Irish music events market, we have calculated the TAM as follows:

- 2024 Music Events Market Revenue: €133.50 million
- Crowd Control and Safety Segment: Estimated at 10% of the total market revenue

TAM Calculation:

- 2024: 10% of €133.50 million = €13.35 million
- 2028: 10% of €143.40 million (with a CAGR of 1.80%) = €14.34 million

The TAM for the Irish crowd control and safety technology market is projected to grow from €13.35 million in 2024 to €14.34 million by 2028.

Serviceable Addressable Market (SAM): PulsePath's cloud-based solution targets a specific segment of the market: real-time, cloud-based crowd control technology. The increasing reliance on digital solutions in the event management sector supports this focus. We estimate that 60% of the total crowd control and safety technology market will be cloud-based.

SAM Calculation:

- 2024: 60% of €13.35 million = €8.01 million
- 2028: 60% of €14.34 million = €8.60 million

The SAM for PulsePath's cloud-based crowd control and safety technology is projected to grow from €8.01 million in 2024 to €8.60 million by 2028.

Serviceable Obtainable Market (SOM): Based on PulsePath's growth potential and competitive positioning, we estimate that the platform can capture 20% of the SAM within its first few years of operation. This projection is realistic given the demand for smarter, data-driven safety solutions and PulsePath's unique value proposition.

SOM Calculation:

- 2024: 20% of €8.01 million = €1.60 million
- 2028: 20% of €8.60 million = €1.72 million

PulsePath's SOM is projected to range from €1.60 million in 2024 to €1.72 million by 2028

Key Insights:

1. **Market Potential:** The Irish crowd control and safety technology market presents significant opportunities, with TAM, SAM, and SOM figures demonstrating a clear growth trajectory.
2. **Cost Efficiency:** PulsePath's potential to reduce security budgets by up to 20% aligns with the market's demand for cost-effective solutions.
3. **Scalability:** By focusing on cloud-based solutions, PulsePath is well-positioned to capture a substantial share of the SAM and grow its SOM over time.

Conclusion: These calculations highlight the substantial market opportunity for PulsePath in the Irish festival and event management sector. By focusing on optimised safety and smarter spending, PulsePath has a clear path to delivering measurable value and capturing a significant share of the market. The platform's scalability and alignment with industry trends position it as a key player in the evolving landscape of event safety and resource management.

Appendix C

ROI Calculation for In-App Advertising

This appendix outlines the detailed calculation of Return on Investment (ROI) for PulsePath's in-app advertising feature, demonstrating its value to food vendors and other festival businesses. By leveraging PulsePath's platform, vendors can gain significant visibility and increased sales, making it an attractive revenue stream for both vendors and PulsePath.

Target Market Analysis:

- Total Addressable Market (TAM):
 - The entire Irish music festival sector, encompassing approximately 238,000 attendees across 11 major music festivals annually.
 - This figure represents the maximum potential user base for PulsePath.
- Serviceable Addressable Market (SAM):
 - Reduced to 119,000 potential users, assuming 50% of festival organisers adopt technology solutions like PulsePath.
- Serviceable Obtainable Market (SOM):
 - PulsePath aims to capture 20% of the SAM, which equates to 23,800 users actively downloading and using the app during festivals in its first year.

ROI Calculation for Food Vendors:

1. Advertising Cost per Vendor:
 - Vendors pay €500 for in-app advertising on PulsePath, including features such as pop-up promotions and highlighted listings to direct foot traffic to their stalls.
2. Impressions and Sales Conversion:
 - Based on PulsePath's SOM of 23,800 users, each vendor advertisement generates 23,800 impressions.
 - With a conversion rate of 10%, this results in approximately 2,380 sales.
3. Revenue Generated per Vendor:
 - Assuming an average transaction value of €12 per sale, vendors achieve total revenue of €28,560 from these sales.
4. ROI Calculation:
 - ROI Formula:
$$\text{ROI} = (\text{Gross Return} - \text{Cost of Investment}) / \text{Cost of Investment}$$

Substituting the values: $\text{ROI} = (\text{€}28,560 - \text{€}500) / \text{€}500 = 56.12\%$

This ROI demonstrates that for every €1 invested in in-app advertising, a vendor earns €0.56 in profit, making PulsePath's advertising model highly lucrative.

Scalability and Revenue Potential:

1. Cumulative Vendor Revenue:
 - Medium to large-scale festivals typically partner with 10 or more food vendors.

- If each vendor achieves similar ROI, the cumulative advertising revenue for PulsePath is substantial, making it a highly scalable revenue stream.
- 2. Reach and Adoption Growth:
 - With a reachable audience of 23,800 users in the first year, more vendors are expected to adopt this model as they recognise the opportunity for increased visibility and revenue compared to traditional methods.

Conclusion: This ROI calculation highlights the compelling value proposition PulsePath provides to food vendors, offering significant revenue potential through targeted advertising. By delivering a high ROI and scalable revenue opportunities, the in-app advertising feature aligns with PulsePath's mission to optimise safety and smarter spending while creating additional value for festival stakeholders.

Appendix D

Post Event Analytics

This appendix summarises the potential for PulsePath to implement a premium post-event data analytics package as an additional revenue stream. By partnering with analytics providers, PulsePath can offer actionable insights that address the growing demand for data-driven solutions in the event management industry.

Key Insights:

1. Rising Demand for Data Analytics:
 - Festival organisers are seeking data-driven insights to improve safety, optimise resource allocation, and enhance attendee experiences.
 - Cloud analytics is not limited to live events; its true value emerges post-event during the evaluation phase. By analysing collected data, organisers can rethink and rearrange strategies based on actionable insights. For instance, assessing attendance patterns and feedback can reveal the popularity of certain activities and help organisers tailor future events to better meet attendee expectations. Feedback on logistical aspects such as registration processes or venue amenities can uncover inefficiencies and guide improvements. (InEvent, 2024)
 - The global predictive analytics market was valued at 5.29 billion U.S. dollars in 2020 and is forecasted to grow to 41.52 billion U.S. dollars by 2028. Predictive analytics are widely used to analyse consumer behaviour, manage supply chains, and optimise business operations, showcasing the growing reliance on data-driven decision-making across industries. (Statista, 2023)
2. Market Opportunity:
 - A premium post-event analytics package offers a big opportunity to cater to organisers looking for long-term budget-friendly measures.
3. Revenue Potential:
 - As data analytics adoption grows, the demand for post-event solutions is expected to grow too, positioning PulsePath to capitalise on this trend and generate significant additional revenue.

The growing interest in data analytics within the event management industry presents an opportunity for PulsePath to establish a scalable revenue stream. By addressing this demand, PulsePath can enhance its appeal to organisers who are looking for data-driven solutions for safety and operational improvements.

References:

- InEvent. (2024, July 5). How to use data analytics in event management. <https://inevent.com/blog/others/how-to-use-data-analytics-in-event-management.html>

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Appendix E

Comparison of Pricing Strategies for Crowd Management Solutions

This table compares PulsePath's pricing model with three similar software solutions, highlighting the key differences and advantages of our approach.

Pricing Model	PulsePath	OnePlan	Sirix	TallyFi
Event-based Pricing	Flat fee per event based on event size, attendee numbers, and facilities requested.	Free for 1 event, then tiered pricing from €22/month.	Scaled pricing based on monitoring requirements, area coverage, and event size. Potential cost-per-event for smaller events or hourly for larger, high-activity zones.	\$50–\$100/month with device limits, billed annually. Custom pricing for larger setups.
In-App Promotions	Commission-based from on-site food vendors and partnerships.	No comparable in-app promotion model.	No in-app promotion model; focused on security and monitoring.	No in-app promotion model; focused on crowd counting and monitoring.
Data Insights	Charge for detailed post-event analytics and reports.	No specific mention of data analytics or post-event reports.	High-level crowd movement and density reports, no detailed post-event analytics.	Weekly summary emails and detailed statistics for each device or location.
Partnership Promotions	Commission from services like	No direct mention of	No direct partnerships but integrates with	No partnerships; focused on

	ride-sharing, hotels, beverage brands, and exclusive merchandise.	such promotional partnerships.	other security systems as needed.	direct crowd analytics for organisers.
Subscription Tiers	No subscription-based pricing; pay-per-use model.	€22–€89/month depending on tier and team size.	Custom pricing for high-level, ongoing remote monitoring with potential hourly rates or cost-per-event options.	\$50–\$100/month per device, billed annually; custom plans for high device usage.

Key Insights:

1. Flexibility for Organisers:
 - Festival organisers often prefer pricing models that don't require long-term commitments. PulsePath's pay-per-use model eliminates the need for financial obligations, making it an ideal choice for cost-conscious organisers.
2. Scalability and Customisation:
 - PulsePath's pricing structure allows organisers to purchase only the features they need, such as post-event analytics, as optional add-ons. This customisation allows organisers to tailor the service to their specific requirements and budgets.
3. Alignment with Industry Trends:
 - The growing popularity of consumption-based pricing in the SaaS industry shows the demand for flexible models. According to OpenView Partners, usage-based pricing (UBP) adoption has nearly doubled in the past five years, with three out of five companies now actively using or testing this model. (OpenView Partners, n.d.) PulsePath's event-based pricing aligns with this trend, providing a modern, customer-focused alternative to subscription-based models.

PulsePath's pay-per-use pricing model is a forward-thinking strategy that addresses the changing demands of festival organisers. PulsePath aligns with trends in B2B SaaS monetisation to provide a flexible, scalable, and cost-effective option, ensuring a strong competitive advantage in the market.

References:

- OpenView Partners. (n.d.). Usage-based pricing: The next iteration of SaaS monetization. OpenView. Retrieved from <https://openviewpartners.com/usage-based-pricing/>

